Abstract

The Multi-Domain Grievance Management System is an intelligent platform designed to streamline the process of managing complaints across multiple domains, including education, healthcare, and public services. The system allows users to register, submit complaints, and track their status throughout the resolution process. Submitted complaints are analyzed using Natural Language Processing (NLP) and Artificial Intelligence (AI) techniques to identify the relevant domain and category, assess severity and priority, and determine the appropriate department. The system also detects duplicate or similar complaints and searches historical complaint records to provide suitable resolution recommendations. Before assignment, administrators can verify and modify the AI-generated results to ensure accurate processing. The complaint is then assigned to the concerned department for resolution, while users can monitor its progress through status updates. User feedback and administrator corrections are collected to support continuous system improvement. The proposed system aims to reduce manual effort, improve complaint classification and routing, minimize delays, and provide an efficient and organized approach to multi-domain grievance management.

1. User Management & Security

FR1: The system shall allow users to register, log in, and log out securely.

FR2: The system shall allow administrators to log in through a separate, role-based access interface.

2. Complaint Submission

FR3: The system shall allow registered users to submit complaints through a web-based form.

FR4: The system shall allow users to attach supporting details or evidence with a complaint.

3. Language Detection & Translation

FR5: The system shall automatically detect the language of a submitted complaint (Telugu, Hindi, or English).

FR6: The system shall automatically translate non-English complaints into English for further processing.

4. Text Preprocessing

FR7: The system shall preprocess complaint text, including cleaning, tokenization, and normalization, before analysis.

5. Domain & Category Classification (NLP/AI)

FR8: The system shall use NLP and AI techniques to detect the relevant domain of a complaint (Education, Healthcare, or Public Service).

FR9: The system shall classify each complaint into an appropriate category within its detected domain.

FR10: The system shall perform context and confidence analysis to validate the accuracy of the classification.

6. Severity, Priority & Location Assessment

FR11: The system shall assess and assign a severity level to each complaint.

FR12: The system shall assess and assign a priority level to each complaint based on severity and context.

FR13: The system shall identify the location associated with a complaint.

7. Department Identification & Assignment

FR14: The system shall identify the appropriate department for a complaint based on domain, category, location, and priority.

FR15: The system shall assign a verified complaint to the concerned department for resolution.

8. Duplicate Detection & Historical Analysis

FR16: The system shall detect duplicate or similar complaints submitted by the same or different users.

FR17: The system shall search historical complaint records related to a new complaint.

FR18: The system shall generate resolution recommendations based on historical complaint data.

9. Administrator Verification

FR19: The system shall allow administrators to review AI-generated classification, severity, priority, and department results.

FR20: The system shall allow administrators to modify or correct AI-generated results before final assignment.

10. Complaint Tracking & Escalation

FR21: The system shall allow users to track the real-time status of their submitted complaints.

FR22: The system shall update complaint status as it progresses through resolution stages.

FR23: The system shall automatically escalate complaints that remain pending beyond a defined time or are marked high-priority.

11. Feedback & System Improvement

FR24: The system shall collect user feedback after complaint resolution.

FR25: The system shall record administrator corrections to support continuous improvement of the AI models.